



The National University of Computer and
Emerging Sciences

Decision Trees

Machine Learning for Data Science

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Goals

- Review of Previous Lecture
- Today's Lecture
 - Decision Trees

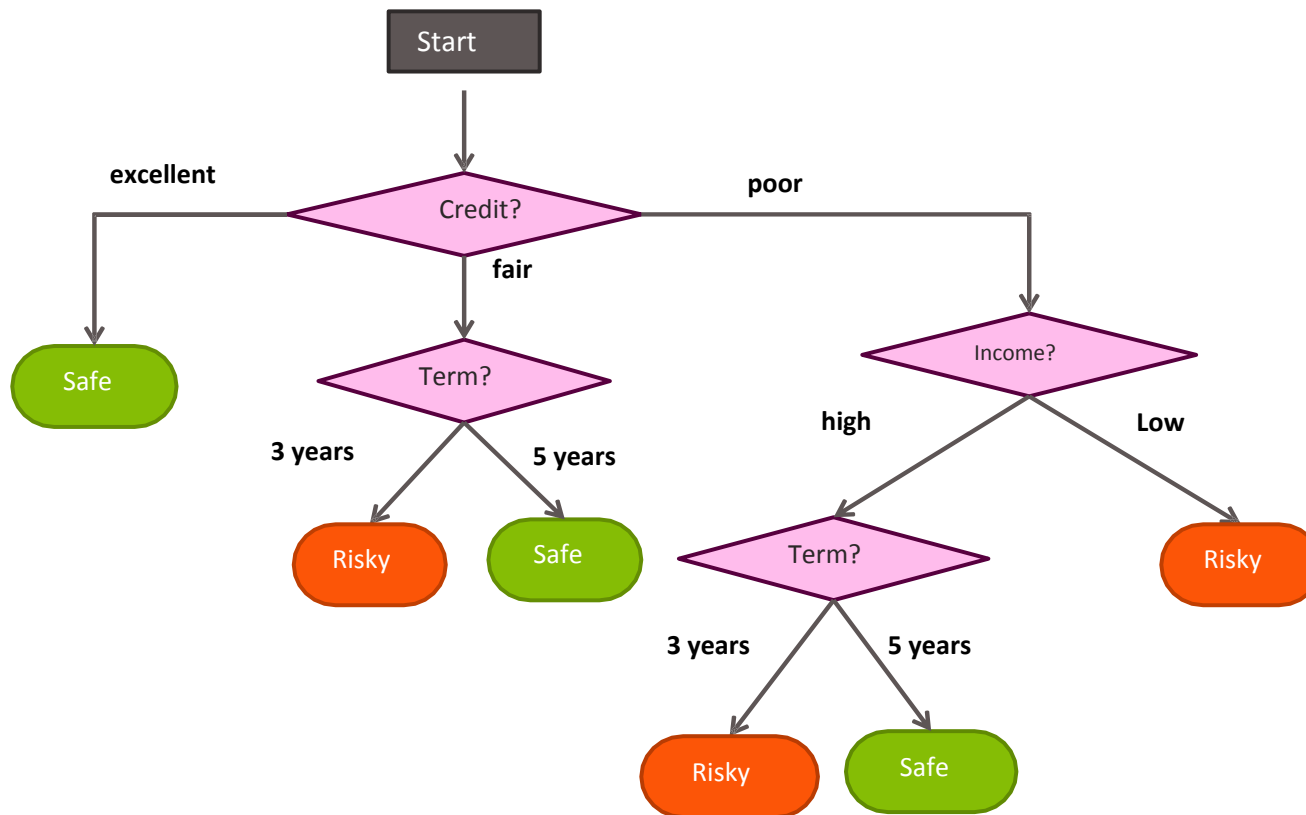
Motivation

- Problem: Given the following information of a customer, decide whether the bank should assign a loan or not
- Classes: Safe or Risky
- Features: Credit History, Income, Term, Personal Info

Credit	Term	Income	y
excellent	3 yrs	high	safe
fair	5 yrs	low	risky
fair	3 yrs	high	safe
poor	5 yrs	high	risky
excellent	3 yrs	low	risky
fair	5 yrs	low	safe
poor	3 yrs	high	risky
poor	5 yrs	low	safe
fair	3 yrs	high	safe

Motivation

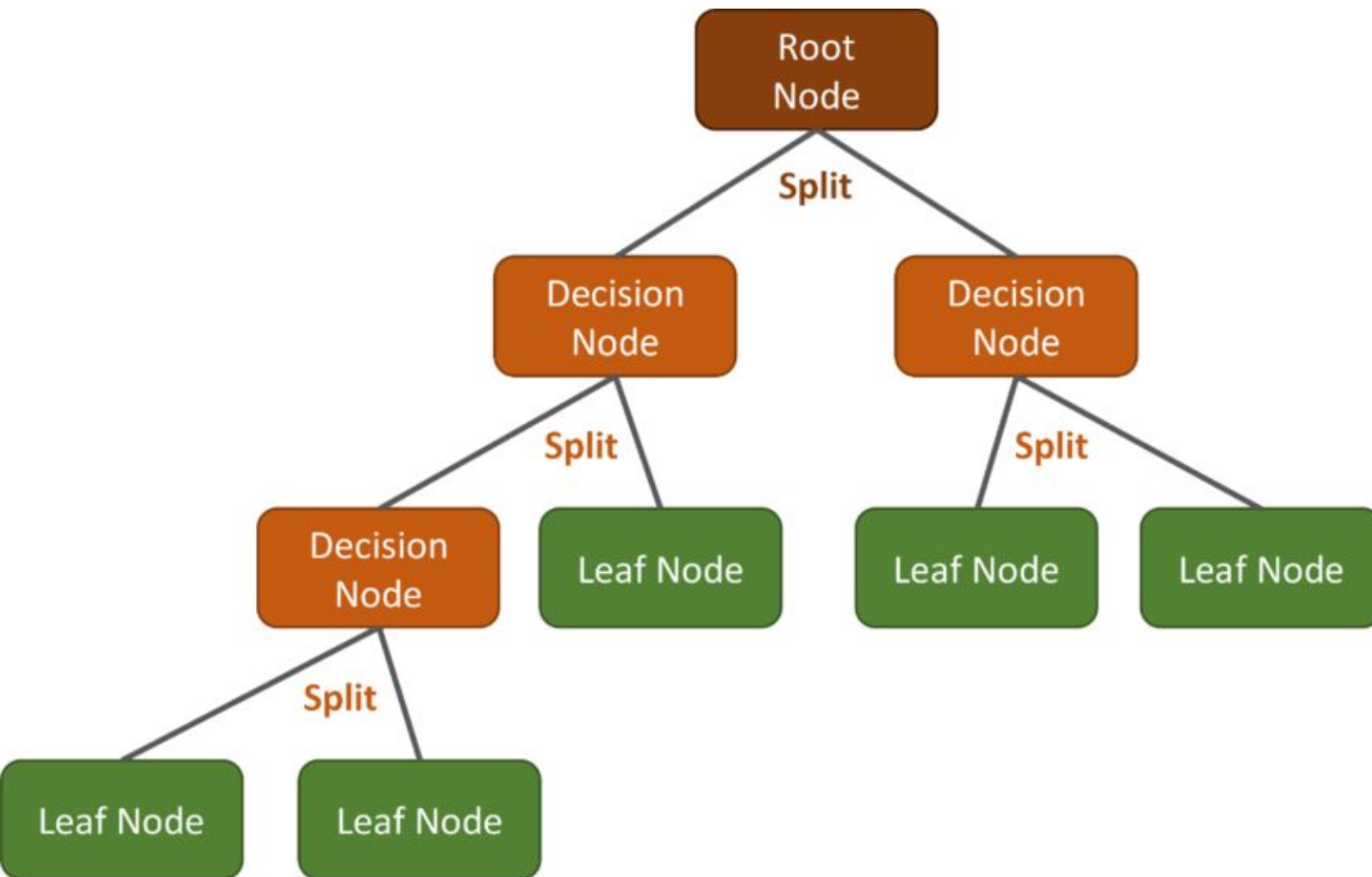
$x_i = (\text{Credit History} = \text{poor}, \text{Income} = \text{high}, \text{Term} = 5 \text{ years})$



Decision Trees

- **Decision tree (DT)** is a machine learning algorithm.
- Its learning function is represented by a **decision tree**.
- DT can also be represented as sets of **if-then rules**.
- Among the most popular of **inductive inference algorithms**
- Applied to a broad range of tasks:
 - Medical diagnosis
 - Credit risk analysis
 - Object classification

Decision Trees



Classification and Regression Trees (CART)

- How do we organize the tree to produce the lowest classification error?
- Any decision tree will successively split the data into smaller and smaller subsets.
 - It would be ideal if all the samples associated with a leaf node were from the same class.
 - Such a subset, or node, is considered pure in this case.
- A generic tree-growing methodology, known as CART, successively splits nodes until they are pure.
 - Classification and Regression Tree

Classification and Regression Trees (CART)

- Six key questions:

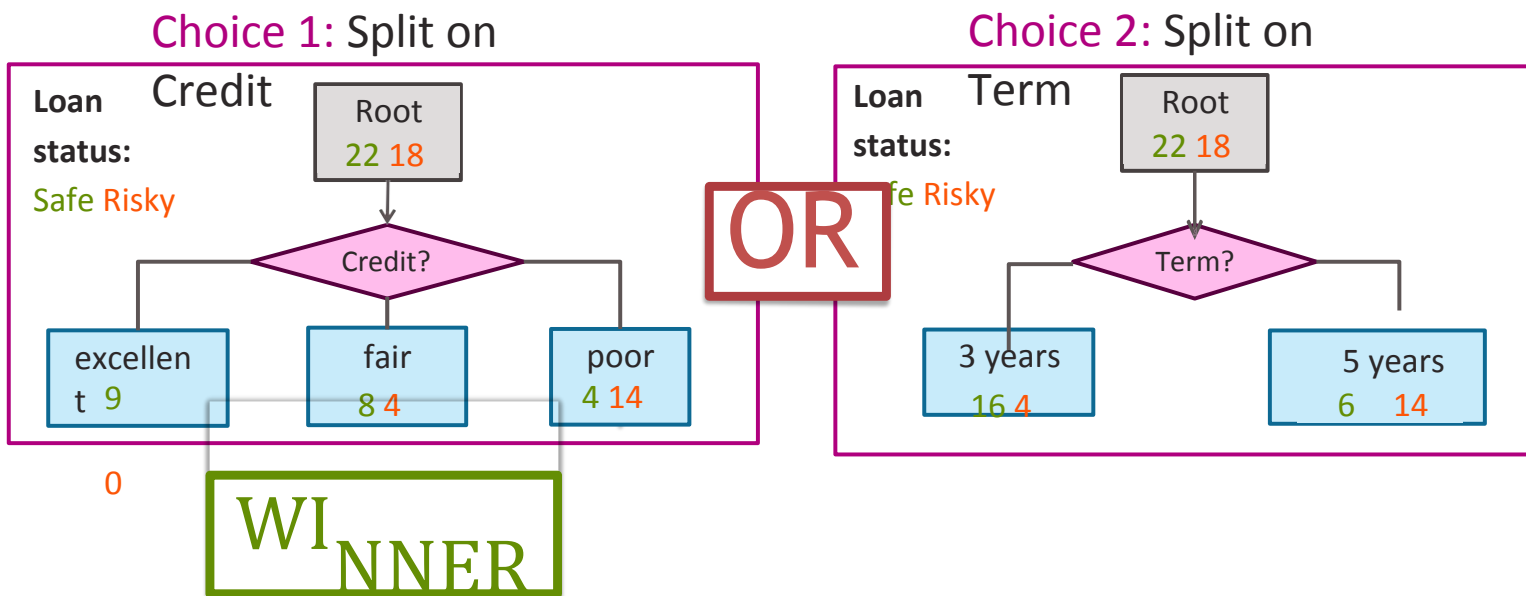
- 1) Should the questions be binary (e.g., is gender male or female) or numeric (e.g., is height $\geq 5'4''$) or multi-valued (e.g., race)?
- 2) Which feature should be tested at each node?
- 3) When should a node be declared a leaf?
- 4) If the tree becomes too large, how can it be pruned?
- 5) If the leaf node is impure, what category should be assigned to it?
- 6) How should missing data be handled?

ID3 Algorithm

- **Iterative Dichotomiser**
 1. $A \leftarrow$ select the "best" decision attribute to split the examples
 2. For each value of A create new child nodes
 3. Split training examples according to the attribute value (Sunny, Overcast, Rain)
 4. If all training examples are perfectly classified stop, else iterate over new leaf nodes.

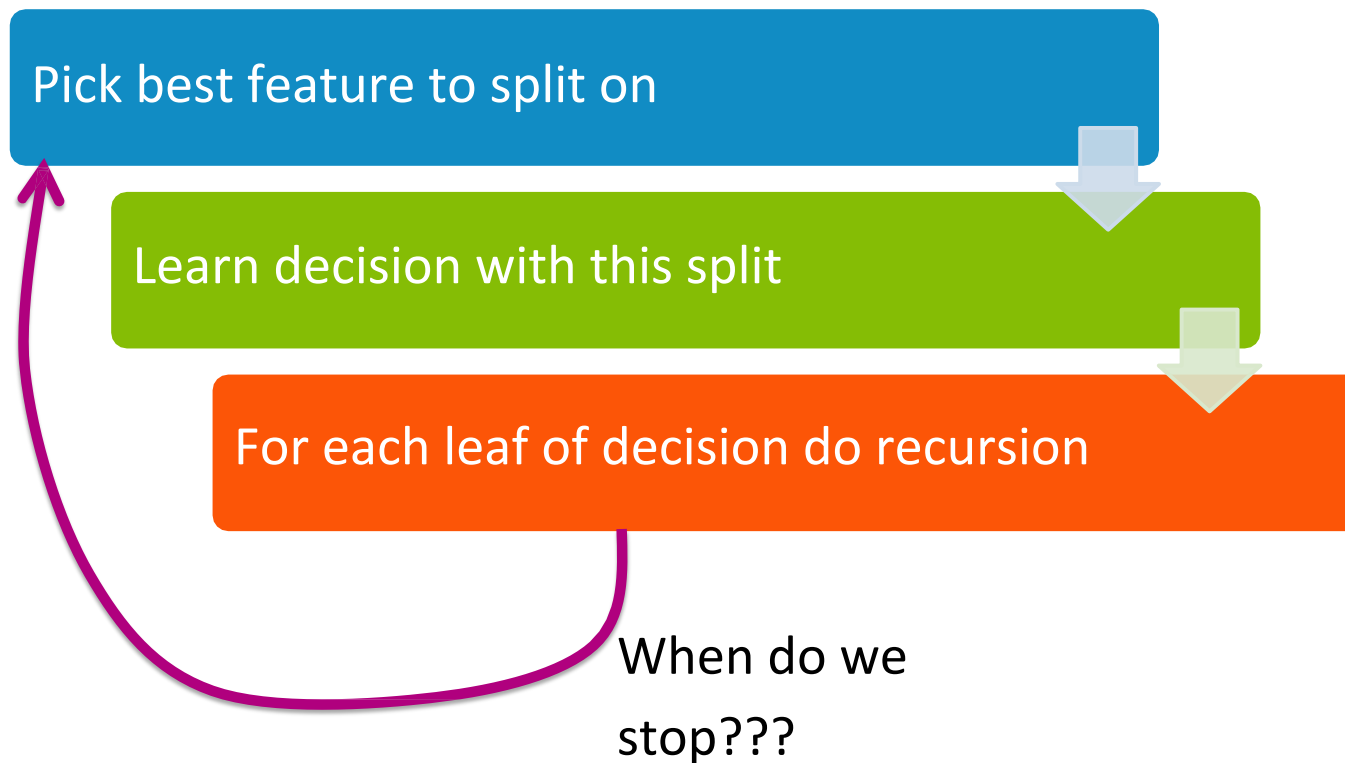
Choice 1 vs Choice 2: Comparing split on Credit vs Term

Tree	Classification error
(root)	0.45
split on credit	0.2
split on loan term	0.25



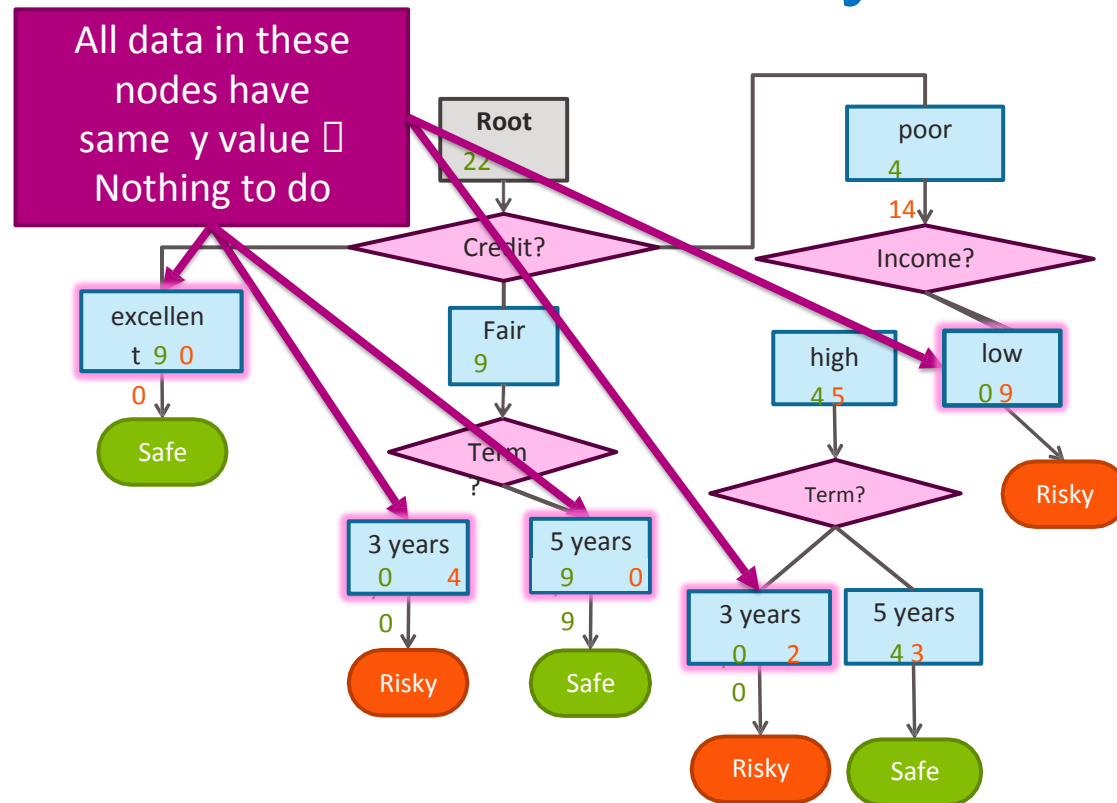
Today's Lecture

Simple greedy decision tree learning

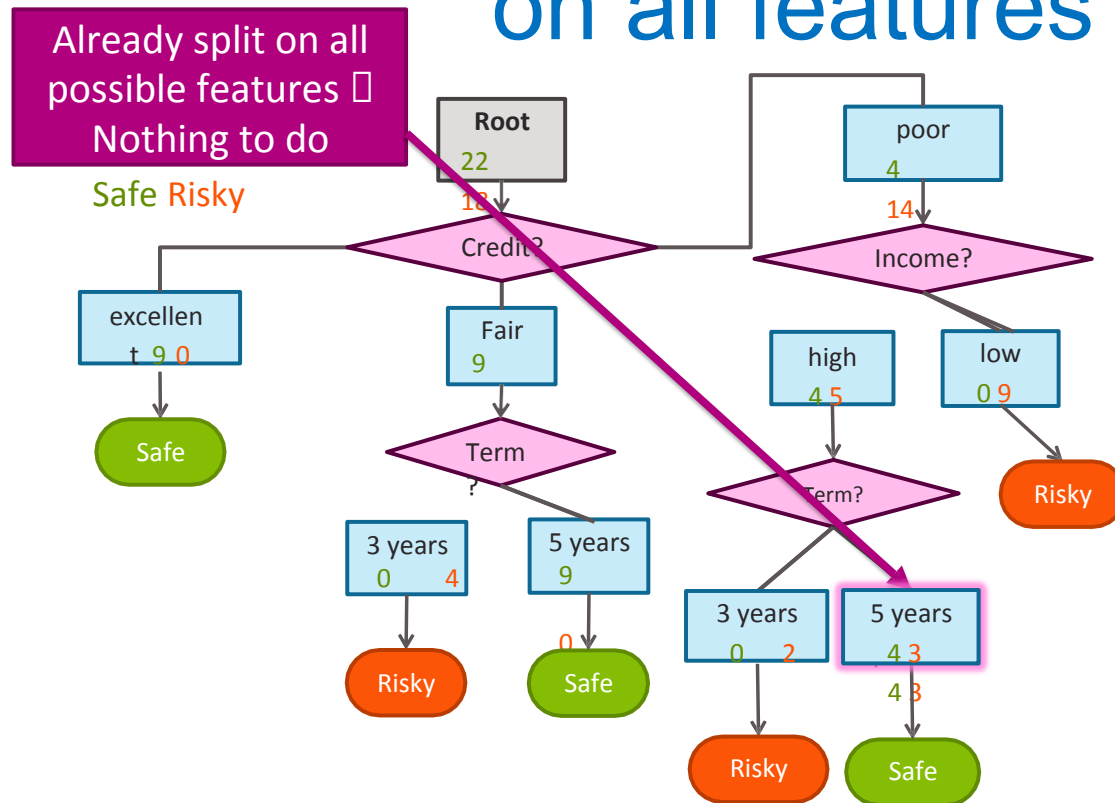


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Stopping condition 1: All data agrees on y



Stopping condition 2: Already split on all features



Greedy decision tree learning

- **Step 1:** Start with an empty tree

- **Step 2:** Select a feature to split data

- For each split of the tree:

- **Step 3:** If nothing more to do, make predictions

- **Step 4:** Otherwise, go to **Step 2** & continue (recurse) on this split

Pick feature split leading to lowest classification error

Stopping conditions 1 & 2

Recursion

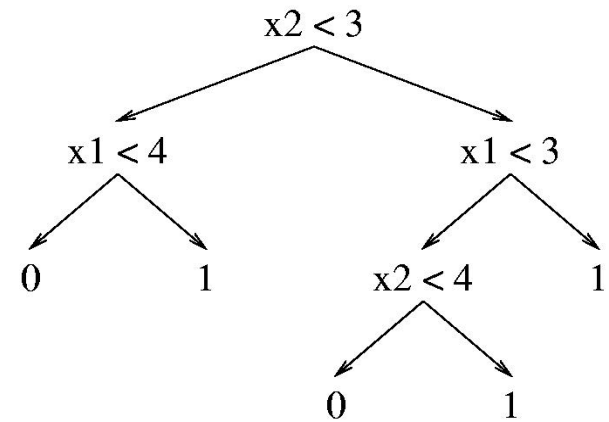
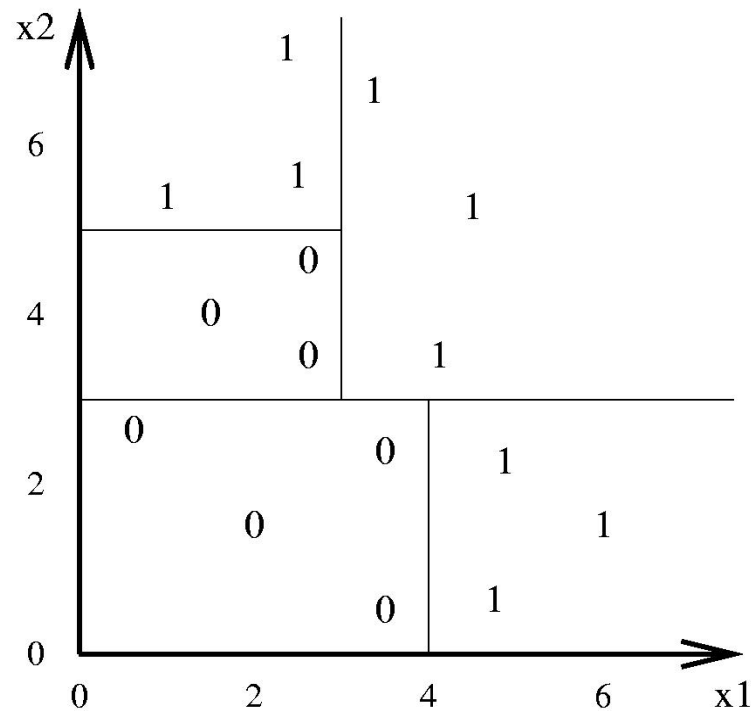
Is this a good
idea?

Proposed stopping condition

3: Stop if no split reduces the
classification error

Decision Tree Decision Boundaries

Decision trees divide the feature space into axis-parallel rectangles, and label each rectangle with one of the K classes.



Decision Trees Provide Variable-Size Hypothesis Space

As the number of nodes (or depth) of tree increases, the hypothesis space grows

- **depth 1** (“decision stump”) can represent any boolean function of one feature.
- **depth 2** Any boolean function of two features; some boolean functions involving three features (e.g., $(x_1 \wedge x_2) \vee (\neg x_1 \wedge \neg x_3)$)
- **etc.**

Learning Algorithm for Decision Trees

The same basic learning algorithm has been discovered by many people independently:

GROWTREE(S)

if ($y = 0$ for all $\langle \mathbf{x}, y \rangle \in S$) **return** new leaf(0)

else if ($y = 1$ for all $\langle \mathbf{x}, y \rangle \in S$) **return** new leaf(1)

else

 choose best attribute x_j

$S_0 =$ all $\langle \mathbf{x}, y \rangle \in S$ with $x_j = 0$;

$S_1 =$ all $\langle \mathbf{x}, y \rangle \in S$ with $x_j = 1$;

return new node(x_j , **GROWTREE**(S_0), **GROWTREE**(S_1))

Choosing the Best Attribute

One way to choose the best attribute is to perform a 1-step lookahead search and choose the attribute that gives the lowest error rate on the training data.

CHOOSEBESTATTRIBUTE(S)

choose j to minimize J_j , computed as follows:

S_0 = all $\langle \mathbf{x}, y \rangle \in S$ with $x_j = 0$;

S_1 = all $\langle \mathbf{x}, y \rangle \in S$ with $x_j = 1$;

y_0 = the most common value of y in S_0

y_1 = the most common value of y in S_1

J_0 = number of examples $\langle \mathbf{x}, y \rangle \in S_0$ with $y \neq y_0$

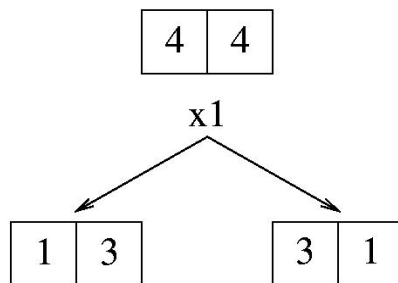
J_1 = number of examples $\langle \mathbf{x}, y \rangle \in S_1$ with $y \neq y_1$

$J_j = J_0 + J_1$ (total errors if we split on this feature)

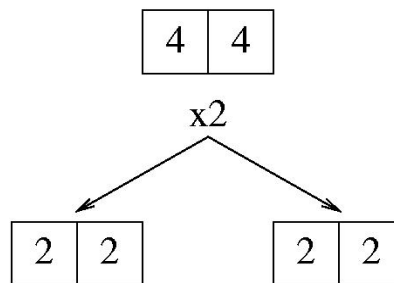
return j

Choosing the Best Attribute—An Example

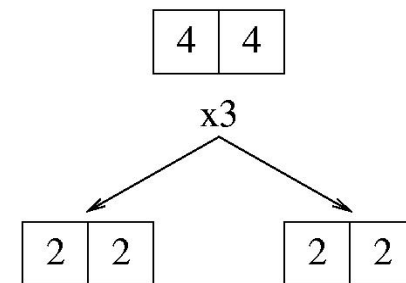
x_1	x_2	x_3	y
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	0



J=2



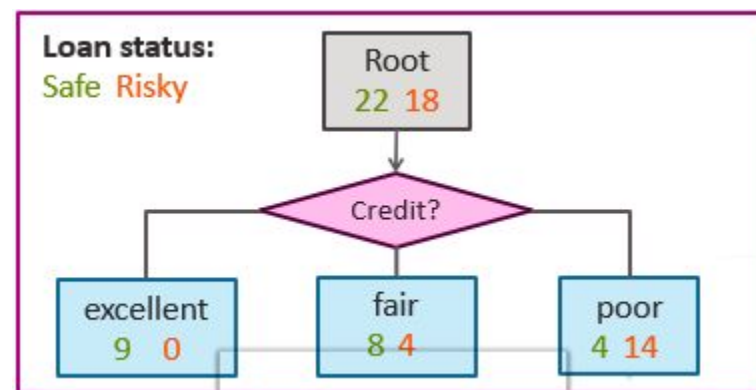
J=4



J=4

Alternate Ways

- The fundamental principle for tree creation
 - Simple and compact tree with few nodes.
- The simplest model that explains data is the one to be preferred
- Find a condition T at each node N that makes the data reaching the immediate descendent nodes as “pure” as possible.
 - Minimum errors
- What is **Pure or impurity**?



Decision Trees

- Different **mathematical measures** of impurity
- The most popular measure is the *entropy impurity*
 - *Information impurity*
- Let $i(N)$ be impurity of a node N .
 - $i(N) = 0$ for pure classes.
 - $i(N) > 0$ for mix classes.

Decision Trees

- The **Entropy** can be calculated as:

$$i(N) = - \sum_j P(\omega_j) \log_2 P(\omega_j)$$

- where $P(\omega_j)$ is the fraction of patterns at node N that are in category ω_j .
 - Entropy is 0, if all samples belong to same class
 - Otherwise it is positive
- **Entropy is maximum** when different **classes are equally likely**.

Decision Trees

- Another definition of impurity is particularly useful in the two-category case.
- It will have **zero impurity** when the node represents **only patterns of a single category**.
- This can be written as

$$i(N) = P(\omega_1)P(\omega_2)$$

- This is called **variance impurity**

Decision Trees

- Variance impurity can be generalized to **two or more categories as:**

$$i(N) = \sum_{i \neq j} P(\omega_i)P(\omega_j) = 1 - \sum_j P^2(\omega_j)$$

- This is called **Gini impurity**
- It give **expected error rate** at node N for random selection of feature.
- It will be **maximum for classes with equal probabilities.**

Decision Trees

- The *misclassification impurity* can be written as

$$i(N) = 1 - \max_j P(\omega_j)$$

- It measures the *minimum probability* that a training pattern would be *misclassified at N*

Decision Trees

- For node N , what **value s should we choose for the property test T ?**
- Choose the test that **decreases the impurity.**
- The decrease in impurity is calculated as:

$$\Delta i(N) = i(N) - P_L i(N_L) - (1 - P_L) i(N_R)$$

- N_L and N_R are the left and right descendent nodes,
- $i(N_L)$ and $i(N_R)$ their impurities
- P_L is the fraction of patterns at node N that will go to N_L

References

- Chapter 3, Machine Learning, Tom Mitchell
- <http://cs229.stanford.edu/notes2021spring>

Thank You 😊